

The Formula Cheat Sheet

For AQA · Edexcel · OCR · WJEC. Since 2022 you get a **formula sheet in the exam** — so the danger isn't the formulas that are printed, it's the ones you **still have to memorise**. Everything below is colour-coded so you know exactly what to commit to memory tonight.

■ Given to you in the exam — don't waste memory on it ■ NOT given — you must memorise this

Trigonometry & non-right-angled triangles

SOH-CAH-TOA

$\sin = O/H$ $\cos = A/H$ $\tan = O/A$
Right-angled triangles only.

MEMORISE

Sine rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

GIVEN

Cosine rule

$a^2 = b^2 + c^2 - 2bc \cdot \cos A$
Angle version: $\cos A = (b^2 + c^2 - a^2) / 2bc$

GIVEN

Area of a triangle

$$\text{Area} = \frac{1}{2} ab \cdot \sin C$$

GIVEN

Exact values

$\sin 30 = 1/2$, $\cos 60 = 1/2$, $\tan 45 = 1$, $\sin 60 = \cos 30 = \sqrt{3}/2$
Could appear; calculator gives them, but know the common ones.

MEMORISE

Pythagoras, circles & mensuration

Pythagoras

$a^2 + b^2 = c^2$
c is the hypotenuse. Also used for 3D diagonals.

MEMORISE

Circle

Area = πr^2 · Circumference = $\pi d = 2\pi r$

MEMORISE

Arc & sector

Arc = $(\theta/360) \times 2\pi r$ · Sector area = $(\theta/360) \times \pi r^2$

MEMORISE

Cylinder / prism

Volume = area of cross-section × length

MEMORISE

Sphere

Volume = $\frac{4}{3} \pi r^3$ · Surface area = $4\pi r^2$

GIVEN

Cone

Volume = $\frac{1}{3} \pi r^2 h$ · Curved SA = $\pi r l$

GIVEN

Algebra & graphs

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

GIVEN

For $ax^2 + bx + c = 0$. Type carefully with brackets on the calculator.

Straight line

$$y = mx + c \cdot \text{gradient } m = \frac{\Delta y}{\Delta x}$$

MEMORISE

Perpendicular lines

$$m_1 \times m_2 = -1$$

Perpendicular gradient = negative reciprocal.

MEMORISE

Proportion

$$\text{Direct: } y = kx \cdot \text{Inverse: } y = k/x$$

Find k from the given pair first, then solve.

MEMORISE

nth term (linear)

$$\text{nth term} = dn + (a - d)$$

d = common difference, a = first term.

MEMORISE

Compound measures & growth

Speed

$$\text{Speed} = \frac{\text{distance}}{\text{time}} \quad (D=S \times T)$$

MEMORISE

Density

$$\text{Density} = \frac{\text{mass}}{\text{volume}}$$

MEMORISE

Pressure

$$\text{Pressure} = \frac{\text{force}}{\text{area}}$$

MEMORISE

Percentage change

$$\frac{\text{change}}{\text{original}} \times 100$$

Reverse %: divide by the multiplier, e.g. $\div 1.2$ to undo a 20% rise.

MEMORISE

Compound interest

$$\text{Amount} = P \times \left(1 + \frac{r}{100}\right)^n$$

Decay: subtract instead of add. Use the power on the calculator.

MEMORISE

Statistics & probability

Frequency density

$$\text{Freq density} = \frac{\text{frequency}}{\text{class width}}$$

Histograms: frequency = FD × class width.

MEMORISE

Mean (frequency table)

$$\text{Mean} = \frac{\Sigma(f \times x)}{\Sigma f}$$

MEMORISE

Tree diagrams

× along branches, + between outcomes

“At least one” = $1 - P(\text{none})$.

MEMORISE

Conditional probability

$$P(A \text{ and } B) = P(A) \times P(B \text{ given } A)$$

Second branch changes if “without replacement”.

MEMORISE

Don't lose the easy marks

- **The formula sheet is in the exam** — but flip to it early so you're not hunting for it under time pressure.
- **Degrees mode** for all trig — check the small **D** on screen.
- **One continuous calculation** with brackets; never round halfway then re-type.

Free to share with anyone sitting Wednesday. You've got this.

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